

RuiwenWANG

AI Infrastructure / ML Systems Engineer — distributed LLM training (5 years at Huawei Paris).
I build automatic-parallelization planners and optimise training performance
— from cost modelling and bottleneck analysis to production integration.



Contact

Paris, France
wangrw0124@gmail.com
(+33) 6 63 11 34 66
French (EU) · open to relocation

Highlights

16,384 NPUs production scale
up to 1.7× faster training
dense · MoE · multimodal models,
up to 1T+ parameters
7 publications · 3 patent families

Core Areas

Training: large-scale LLM pretraining ·
MoE · long context
Parallelism: DP / TP / PP / EP · ZeRO-style
sharding · pipeline scheduling ·
recomputation
Planning: cost models (compute,
memory, communication) · ILP
optimisation · graph (IR) level
Performance: profiling · bottleneck
analysis · tuning

Technical Skills

Languages: Python, C++
Frameworks: PyTorch, Megatron-LM,
MindSpore, MindFormers
Systems: Ascend CANN,
HCCL (NCCL-style collectives)
Tools: Linux, Git, ILP solvers

Awards

Best Poster Award — Euro-Par 2025
Golden Prize — Huawei, 2024
Letter of Appreciation — China Telecom,
2025

Languages

English · French · Mandarin · Cantonese

Online

ruiwen.wang
linkedin.com/in/ruiwen-wang-ai
Google Scholar

EXPERIENCE

Ph.D. Researcher (CIFRE) — LLM Training Systems

2023 – 2026

Huawei Paris Research Center · Sorbonne Université & EURECOM · Paris

- Built automatic-parallelization planners combining compute, memory and communication cost models with optimisation; integrated into MindSpore / MindFormers and benchmarked against Megatron-LM.
- Enabled production use through configuration guidance, debuggable traces and regression benchmarks across model families and cluster scales.

Research Apprentice → Research Engineer — AI Infrastructure

2021 – 2023

Huawei Paris Research Center · Distributed & Parallel Computing Lab · Paris

- Profiled and tuned large-scale LLM training and inference on Ascend-910 production clusters, identifying compute, memory and communication bottlenecks.

DEPLOYMENTS & IMPACT

- Scale:** Ascend-910 clusters up to **16,384 NPUs** — dense, MoE and multimodal models up to 1T+ parameters (DeepSeek, Llama, Mistral / Mixtral, Qwen).
- Production training:** supported China Telecom's DeepSeek-MoE deployment (Letter of Appreciation) and TeleChat3-MoE training.
- Portability:** cost surfaces separate model structure from hardware rates, enabling plans to retarget across clusters and accelerator families without re-profiling.

PUBLICATIONS

First-author publications

- Concord** — NeurIPS 2026. Dependency-keyed incremental re-planning; **~2.7× median re-planning speedup** over full search, with identical plans across 54 audited cases.
- PRISM** — SCA/HPC Asia 2026. Profiling-free symbolic compute, memory and communication costs; **4.2–8.0% mean absolute percentage error** for peak memory.
- BMPipe** — IEEE CLUSTER 2025. Joint pipeline bubble-memory optimisation; **1.7× throughput** with interleaving vs non-interleaved 1F1B on Llama-175B at **16,384 NPUs**.
- H²O** — Euro-Par 2025, Best Poster Award. Joint parallelism search; **+37% / +22% throughput** vs automated / expert plans on DeepSeek-141B (128 NPUs).
- ManuMatic** — IFIP NPC 2025. Pin critical operator shardings and automate the rest; up to **2.4× faster expert computation** on Mixtral-8×7B vs the baseline plan.

Co-authored publications

- MLLM Pipeline Bubble Modeling** — SCA/HPC Asia 2026 (poster).
- TeleChat3-MoE** — arXiv 2025 (technical report).

PATENTS

- Execution-plan generation** — WO 2025/020165 A1 (PCT), Huawei; published 2025.
- Large-model load balancing** — Huawei; pending publication.
- Multidimensional parallelism configuration** — Huawei; pending publication.

EDUCATION

Ph.D. Computer Science (CIFRE) — Sorbonne Université

2023 – 2026

Defended 24 July 2026 · Research with EURECOM and Huawei Paris.

M.Sc. Computer Science — Sorbonne Université

2019 – 2022

B.Sc. Computer Science — Université Paris-Saclay

2016 – 2019